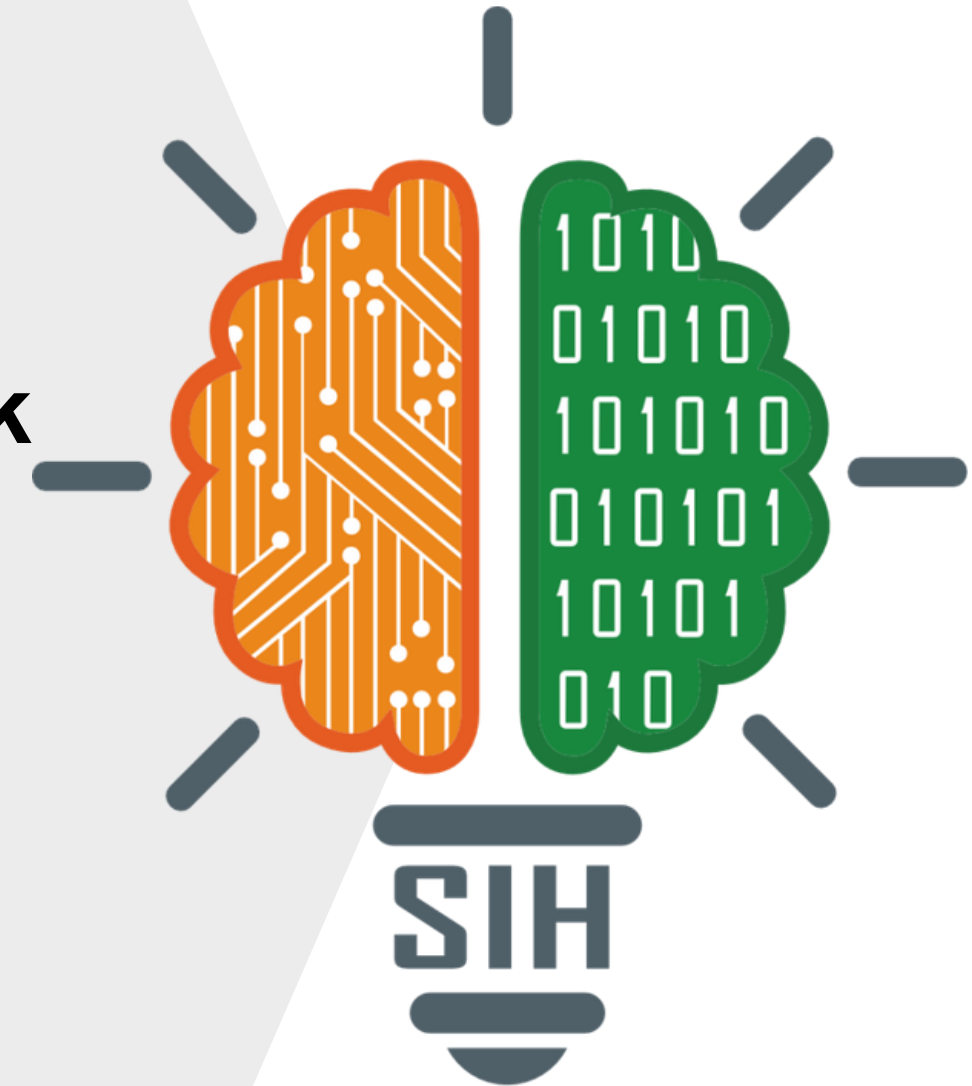


TITLE PAGE

- Problem Statement ID – **SIH26155**
- Problem Statement Title – **AI-Driven Multi-Vendor Network Security Compliance Auditor**
- Theme – **Blockchain & Cybersecurity**
- PS Category – **Software**
- Team ID –
- Team Name (Registered on portal) – **TECHNOOVA**



NEXAUDIT AI

AI-Assisted Multi-Vendor Network Security Compliance Auditor

Proposed Solution

AI-assisted *configuration normalization*

Converts vendor syntax → *common security representation*

Maps configurations → *security controls*

Evidence-linked compliance findings

Confidence-based AI interpretation

Human-verified learning repository

How It Addresses the Problem

Removes the *vendor-syntax barrier*

Creates *one common security language*

Separates AI interpretation from compliance validation

Makes findings *traceable and explainable*

Reuses verified mappings in future audits

Innovation & Uniqueness

Common Security Representation (CSR)

AI + rule-engine hybrid

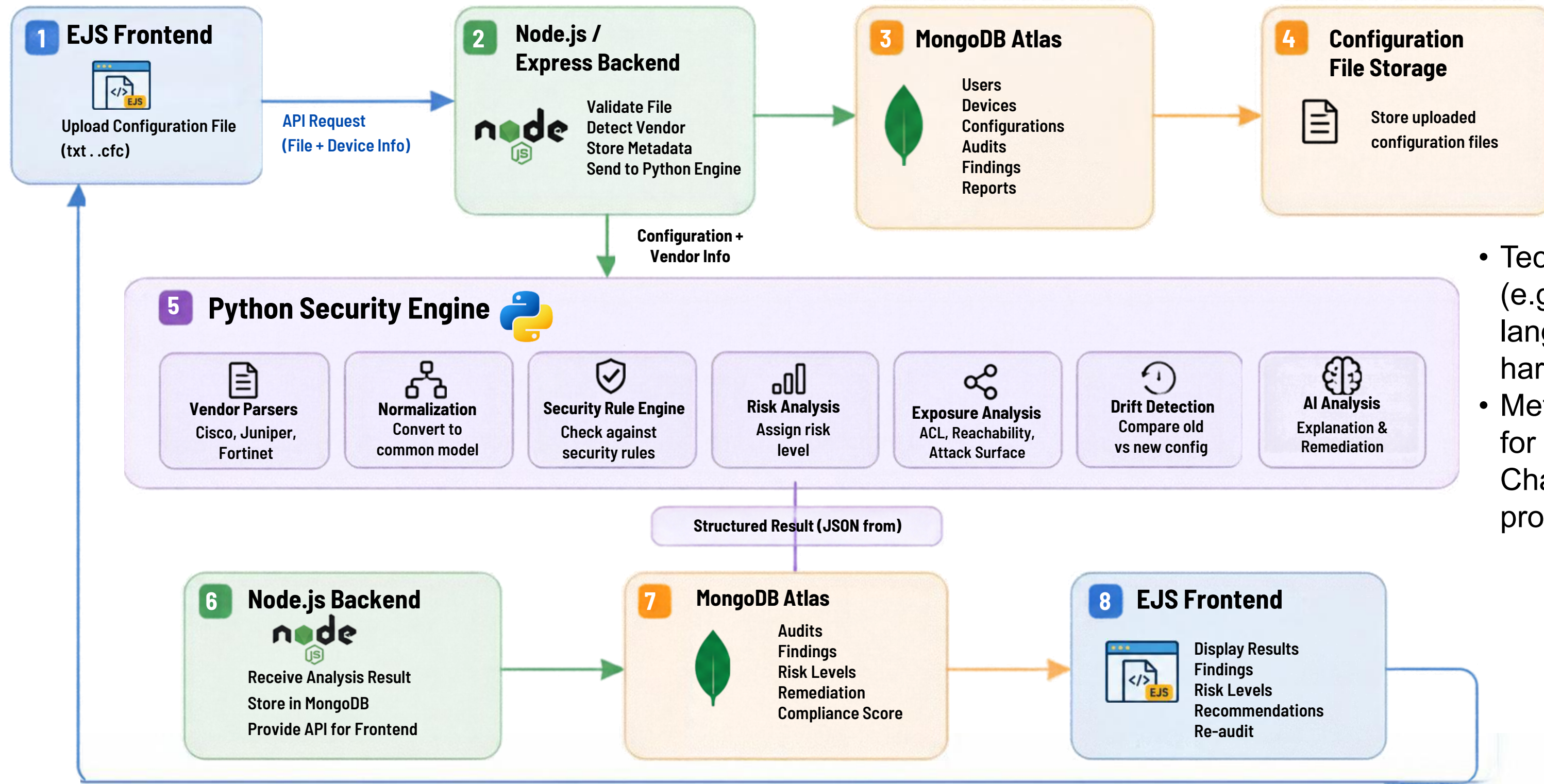
Human-in-the-loop learning

Confidence-aware AI

Vendor-agnostic compliance layer

Evidence-first auditing

TECHNICAL APPROACH



- Technologies to be used (e.g. programming languages, frameworks, hardware)
- Methodology and process for implementation (Flow Charts/Images/ working prototype)

- Analysis of Feasibility**

- File-based MVP* — no live-device access required
- Modular architecture* — easy to extend
- Hybrid AI + Rules* — reliable validation
- API-ready design* — supports future integrations
- Scalable knowledge base* — new mappings can be added
- Cloud deployable* — suitable for centralized auditing

- Potential Challenges & Risks**

- Vendor syntax variations
- Unknown configuration commands
- AI interpretation errors
- Large configuration files
- Changing compliance standards
- Sensitive configuration data

- Strategies to Overcome**

Challenge	Strategy
Syntax variations	Vendor-specific parsers + normalization
Unknown commands	AI + confidence score + admin verification
AI errors	Rule engine for final validation
Large files	Chunked / structured parsing
Standard changes	Version-controlled rule sets
Sensitive data	Encryption + RBAC + secure storage

IMPACT AND BENEFITS

- **Impact on Users**

Security Teams → One view of all vendors

Network Admins → Easier issue identification

Auditors → Clear audit evidence

Organizations → Consistent security checks

- **Future Impact**

Add more network vendors

Add more compliance frameworks

Enable continuous monitoring

Integrate with live network devices

Support organization-wide security dashboards

- **Key Benefits**

Reduces repetitive manual checking

Gives *prioritized security findings*

Keeps *evidence with every finding*

Reuses verified command mappings

Makes audits easier to track

Supports different security frameworks

Simplifies comparison across vendors

RESEARCH AND REFERENCES



- **Research Sources**

NIST Cybersecurity Framework — Security guidelines

CIS Controls — Security best practices

Cisco Documentation — Network configuration reference

Fortinet Documentation — Network configuration reference

Palo Alto Documentation — Network configuration reference

- **References**

NIST Cybersecurity Framework

CIS Controls

Cisco Security Documentation

Fortinet Documentation

Palo Alto Networks Documentation

- **Research Areas**

Network Security

Configuration Auditing

Compliance

AI-assisted Analysis

THANK YOU!